SUPPORTING ADULT GUIDANCE

Meet the parabola

Quadratics. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson introduces the **graph** of a quadratic — the third face of the object the learner has met as a sequence and as an expression. The rule is the simplest one, $y = x^2$. The whole lesson is built on a single honest method: make a **table of values** (substitute each x, square it, get y), **plot** the points, and **join** them into a curve. The key insight — and the reason the graph is a symmetric U rather than a rising line — is that **squaring a negative gives a positive**, so $x = –3$ and $x = 3$ both give y = 9. That produces the **line of symmetry** (here $x = 0$) and the **turning point** (here the minimum, (0, 0)). Stretching, sliding and reflecting this curve — $y = ax^2$ and $y = (x – p)^2$ — are the next two lessons; today stays deliberately on $y = x^2$ so the shape and its symmetry are secure first.

Driving Question: *Why does $y = x^2$ bend into a U?*

The mathematics, in plain terms

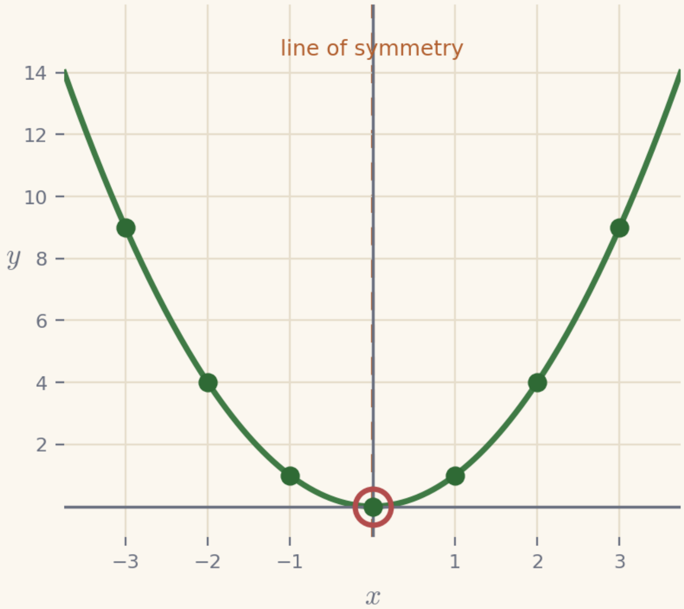
A **parabola** is the U-shaped curve you get when you plot a quadratic. The learner already knows how to make a graph from a rule: choose some x-values, work out the matching y-values, plot the points and join them. Applied to **$y = x^2$** the only new thing to watch is the **negative inputs**. Squaring removes the minus sign — $(–3)^2 = 9$, exactly the same as $3^2 = 9$ — so the left-hand side of the table is a mirror image of the right. That is why the curve does not keep falling on the left but turns and climbs again, giving the symmetric U. The vertical line the curve mirrors across is the **line of symmetry** (for $y = x^2$ it is the y-axis, $x = 0$), and the point where the curve turns is the **turning point** (here the lowest point, (0, 0)). The most useful thing you can do while supporting is to insist on **squaring the negative correctly** — a very common slip is to write $(–3)^2 = –9$. If the learner sees that both 3 and –3 land at height 9, the symmetry, and the whole shape, follow.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — build the parabola from a table

Make a table for y = x^2, plot the points, and join them. Substitute each x and square it.

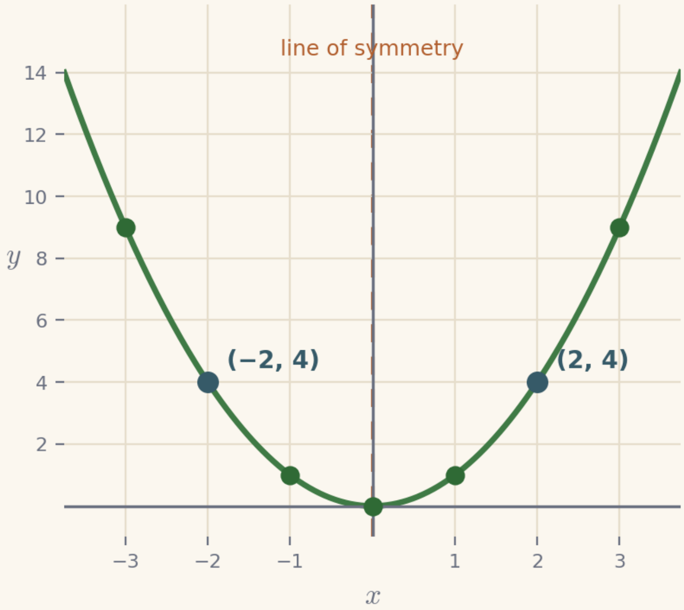


1. $x = -3$ $(-3)^2 = 9$
2. $x = -1$ $(-1)^2 = 1$
3. $x = 0$ $0^2 = 0$
4. $x = 2$ $2^2 = 4$

the seven points make a symmetric U — a parabola — with line of symmetry $x = 0$ and turning point (0, 0)

Example 2 — find the mirror twin

The point (2, 4) is on $y = x^2$. Find its mirror twin — the other point at the same height.



1. the point given	(2, 4)
2. its height is	$2^2 = 4$
3. a twin at height 4 needs	$x^2 = 4$
4. the other value is	$x = -2$, since $(-2)^2 = 4$

the mirror twin is (–2, 4); the two mirror across the line of symmetry $x = 0$

Substitute, plot, then look for the mirror

Three moves carry this lesson. First, **substitute and square**: for each x work out $y = x^2$, being careful that a negative squared is positive. Second, **plot** the points from the table — one at a time is fine — and only then **join** them into a smooth curve, so the learner sees the U emerge from the points rather than being handed it. Third, **look for the mirror**: once a few points are down, ask "is there another point at the same height?" so the line of symmetry is something the learner notices, not a term to memorise. Keep the whole lesson on $y = x^2$; resist the urge to introduce a-stretches or p-slides yet (they are the next two lessons). The stretch — plotting half-way points such as 1.5 — is worth reaching only once the whole-number curve is secure; its job is to show the curve is **continuous**, with a height for every x, not just the whole numbers the sequence used.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing

Connection	Build the curve from a table: substitute y = x^2 for x = –3…3, plot, and join into a parabola.	~12 min
Movement	Predict the height: choose an x, work out y = x^2, and watch the point land on the curve.	~10 min
Reflection	The mirror: find a point's twin across the line of symmetry; notice –a and a share a height.	~10 min
Creativity	Between the whole numbers (stretch): plot a half-way x and see the curve is smooth everywhere.	~9 min
Rest	A pause after watching a rule become a shape.	~2 min
Nutrition	Mode B (intellectual): what the learning feeds — a taste for symmetry, and the confidence that mathematics is connected.	~2 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Squaring a negative to a negative: writing $(–3)^2 = –9$

Repair: “A square is a number times itself. (–3) × (–3): negative times negative is positive, so $(–3)^2 = 9$, the same as 3^2.”

Expecting the curve to keep falling on the left (like a straight line)

Repair: “Check the heights: at $x = –1$ it is 1, at –2 it is 4, at –3 it is 9 — the left side climbs. That is why it is a U, not a line.”

Joining the points with straight segments

Repair: “The curve is smooth. Plot a half-way point like $x = 1.5$ (y = 2.25) — it sits between the dots, so the join must curve.”

Thinking the turning point is just ‘where you started plotting’

Repair: “The turning point is the lowest point, (0, 0). It is the smallest x^2 can be, because a square is never negative.”

Reading the line of symmetry as a horizontal line

Repair: “It is vertical — it runs up and down through the turning point. For $y = x^2$ it is the line $x = 0$.”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: substituting into a rule, squaring a number including squaring a negative, and plotting coordinates. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Keep the learner substituting, plotting and looking for the mirror:

- “Work out y for x = –3, –2, –1, 0, 1, 2, 3. What do you notice about the two ends?”
- “$(-4)^2$ — is that positive or negative? Why?”
- “The point (2, 4) is on the curve. Is there another point at height 4? Where?”
- “Stretch: what is $(1.5)^2$? Where does that point sit compared with (1, 1) and (2, 4)?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“Why is the graph a U and not a rising line?”
Because squaring a negative gives a positive. The negative x-values give the same heights as the matching positive ones, so the left arm mirrors the right and the curve turns at the bottom.

“What are the line of symmetry and the turning point?”
The line of symmetry is the vertical line the curve mirrors across — for $y = x^2$ it is $x = 0$. The turning point is where the curve turns; here it is the lowest point, (0, 0).

“Do they need to know $y = ax^2$ or $y = (x - p)^2$ yet?”
No — those are the next two lessons. Today stays on $y = x^2$ so the shape, the symmetry and the turning point are secure before the curve is stretched or slid.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Emerging → Developing → Secure → Mastering across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Emerging	Beginning to engage; names some features and makes a start with support.
Developing	Carries out the method with prompts; can explain part of the reasoning.
Secure	Works through independently and explains why each step is true.

Mastering	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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